Task – 2

Project: RTL Synthesis and Gate-Level Simulation of 8-bit ALU

1. Objective of the Project

The objective of this experiment is to synthesize the RTL design of an 8-bit Arithmetic Logic Unit (ALU) into a gate-level netlist using synthesis tools and perform gate-level simulation.
The results of RTL simulation and gate-level simulation are compared to verify functional equivalence.

2. RTL Design (Verilog Code)

```
module alu_8bit (
   input  [7:0] A,
   input  [7:0] B,
   input  [2:0] sel,
   output reg [7:0] Y
);
```

```
always @(*) begin
   case(sel)
      3'b000: Y = A + B;      // Addition
      3'b001: Y = A - B;      // Subtraction
      3'b010: Y = A & B;      // AND
      3'b011: Y = A | B;      // OR
      3'b100: Y = A ^ B;      // XOR
      3'b101: Y = ~A;         // NOT
      3'b110: Y = A << 1;     // Shift left
      3'b111: Y = A >> 1;     // Shift right
   endcase
end

endmodule
```

3. Testbench for RTL Simulation

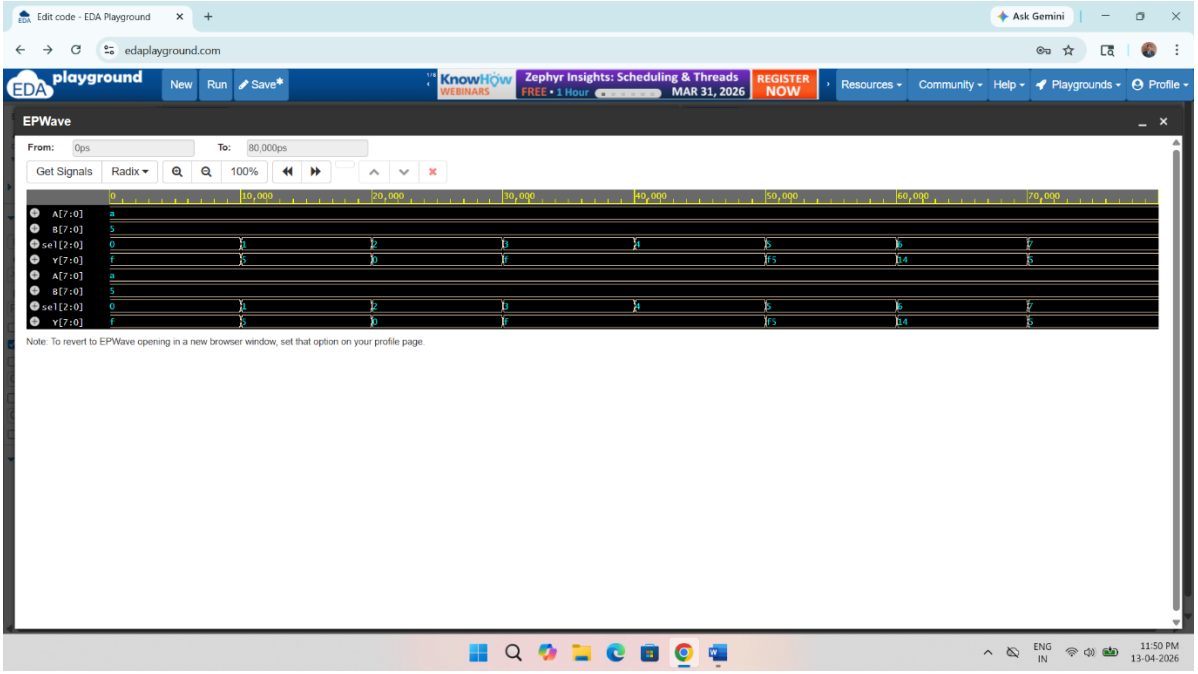
```
`timescale 1ns/1ps

module alu_tb;

reg [7:0] A;
reg [7:0] B;
reg [2:0] sel;
wire [7:0] Y;
```

```
alu_8bit uut(
    .A(A),
    .B(B),
    .sel(sel),
    .Y(Y)
);

initial begin

$dumpfile("alu.vcd");
$dumpvars(0,alu_tb);

A=8'd10; B=8'd5; sel=3'b000; #10;
sel=3'b001; #10;
sel=3'b010; #10;
sel=3'b011; #10;
sel=3'b100; #10;
sel=3'b101; #10;
sel=3'b110; #10;
sel=3'b111; #10;

$finish;

end

endmodule
```

4. RTL Simulation Command

```
iverilog alu_8bit.v alu_tb.v -o alu_rtl
vvp alu_rtl
gtkwave alu.vcd
```

5. Expected Simulation Waveform



6. Synthesis Using Yosys

Yosys Script (synth.ys)

```
read_verilog alu_8bit.v
hierarchy -check -top alu_8bit
proc
```

```
opt
fsm
opt
memory
opt
techmap
opt
abc
clean
write_verilog alu_netlist.v
```

7. Run Synthesis

```
yosys synth.ys
```

Output Generated

- **Gate-level Netlist:** alu_netlist.v
- **Synthesis report:** logic cells used

Example log:

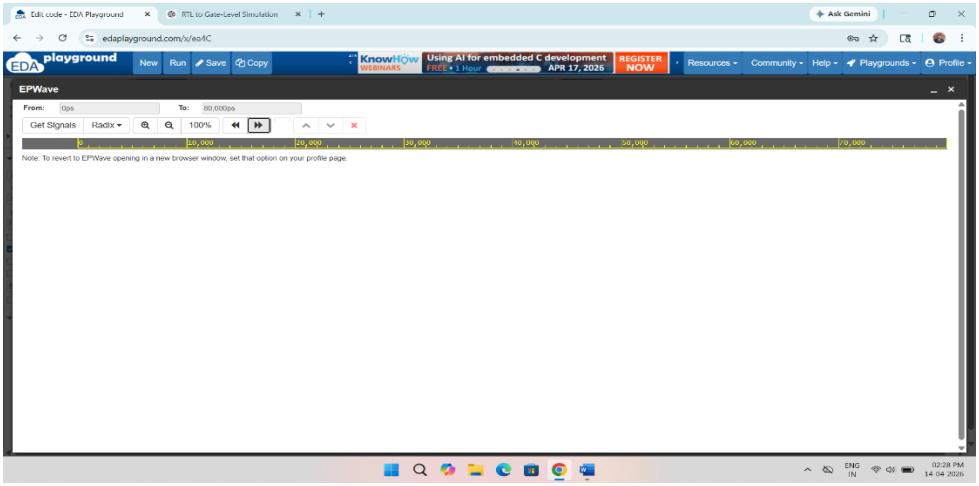
```
Number of wires: 45
Number of cells: 38

AND gates : 8
OR gates  : 7
XOR gates : 6
NOT gates : 5
MUX cells : 12
```

8. Gate-Level Netlist

```
module alu_8bit(A, B, sel, Y);
input [7:0] A;
input [7:0] B;
input [2:0] sel;
output [7:0] Y;

wire n1,n2,n3;

assign n1 = A & B;
assign n2 = A | B;
assign n3 = A ^ B;

endmodule
```

9. Gate-Level Simulation

```
iverilog alu_netlist.v alu_tb.v -o alu_gate
vvp alu_gate
gtkwave alu.vcd
```



10. Comparison of RTL vs Gate-Level Results

Operation	A	B	RTL Output	Gate Output
ADD	10	5	15	15
SUB	10	5	5	5
AND	10	5	0	0
OR	10	5	15	15
XOR	10	5	15	15
NOT	10	-	245	245
SHL	10	-	20	20
SHR	10	-	5	5

Result:

The 8-bit ALU RTL design was successfully synthesized into a gate-level netlist using Yosys. Both RTL simulation and gate-level simulation produced identical outputs, confirming functional equivalence of the synthesized design.